Information Retrieval and Extraction : Lecture I

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Information Retrieval and Extraction

Technical Details

- References
  - Introduction to Information Retrieval, Christopher Manning et al.
  - If I follow other references for a particular module, will refer to that
- Grading Schema
  - MidSem : 30 %, End Sem : 40 %, Project : Group or Individual 2 quizzes
- Focus on application oriented wrt concepts
- It's a practical subject, so for actual knowledge, strongly suggest for group projects
  - will also try for small implementation based observations for some classes at regular intervals

Introduction

Overview

- Hopefully, an interesting course
  - All of us are completely dependent on internet
- Continuous influx of new information
  - Easy availability of information
  - How often have you seen celebrities fall into their own trap based on their history on social media platforms?
- I am not telling, this is the greatest use of internet
  - Highlighting the ability of a stranger as compared to any nosy neighbor
  - Digging through enough information to get to the **most relevant** information

Introduction

Definition

Information retrieval (IR) is **finding material** (usually documents) of **unstructured** nature (usually text) that satisfies an **information need** from within large collections (usually stored on computers)

Examples

- Remember librarians, efficient ones!
- Paralegals!
- The world has changed,
  - Hundreds of millions of people engage in information retrieval!
  - Finding the right email to finding the profile, finding the complete information
  - Observation outsourcing even information retrieval

Introduction

Definition

- What is the huge cry behind **unstructured data**?
  - Find the person with this roll number
  - Transfer the money to this account number
  - Structured, through **indexing**
    - refers to data which does not have clear,semantically overt, easy-for-a-computer structure
  - Is text structured?
    - what about any language?
  - Filtering document collections or further processing a set of retrieved documents
    - Clustering or Classification?
  - Given a set of pre defined topics, we allocate each book to one of the topics
    - is this classification or clustering?
    - what happens for a new book which does not match with any?
    - basic difference between classification and clustering in this case is?

Introduction

Motivation

- Major challenge lies in the scale of the data for information retrieval
  - identifying the relevant information wrt query or information need from a small set of sources is not that complex
- Find all the books by Shakespeare which has **Brutus** and **Caesar** but not **Caliphurnia**
  - linearly scan all the documents and search for each of these names
  - one that satisfies is our **results**!
- Challenges
  - To process large document collections quickly
    - How much do you wait for your response!
  - To allow more flexible matching operations
    - they must co occur
  - Ranked retrieval
    - I need the best possible **response** to my **query**

Introduction

Motivation

- Suppose, you are in trivia and your partner gives an hint
- Possible answer space
  - Kuch Kuch Hota Hai, Dil To Pagal Hai, Kabhi Khushi Kabhie Gham, Darr, Yes Boss, Zamaana Deewana, Har Dil Jo Pyaar Karega, and Chennai Express
- Which is the most relevant based on **hint**?
- if *Har Dil Jo Pyaar Karega* was the answer, how likely will you swear to yourself to change partner forever?

Solution

- Avoid linear scanning, formulate indexing!
- Let us stick with Shakespeare's Collected Works
  - Create a binary term-document incidence matrix
  - what is a term? what is a document?

Boolean Retrieval

Representation

- Break the **document** into distinct words or tokens or items or terms
  - What about punctuations?
  - Capitalized letters?
  - Same words, different forms?
  - Same meaning, different words?
  - All words we keep of all forms?
  - Numbers or hashtags or numbers combined with letters?
- Create the incidence matrix
  - rows represent each word
  - columns represent each document

Boolean Retrieval

Representation

	Antony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth	...
Antony	1	1	0	0	0	1	
Brutus	1	1	0	1	0	0	
Caesar	1	1	0	1	1	1	
Calpurnia	0	1	0	0	0	0	
Cleopatra	1	0	0	0	0	0	
mercy	1	0	1	1	1	1	
worser	1	0	1	1	1	0	
...							

Boolean Retrieval

Methodology

- To answer the query *Brutus AND Caesar AND NOT Calpurnia*
  - Take the corresponding vectors for each required term or token
  - Apply the boolean operation
  - 110100 AND 110111 AND 101111 = 100100
  - Therefore, the result is **Antony and Cleopatra** and **Hamlet**
- Boolean Retrieval means
  - pose any query which is in the form of a Boolean expression of terms
  - tokens connected by boolean operators, AND, OR, NOT
  - view a document as a collection of words
- Challenges
  - when was the last time you queried in **Boolean Operator** ?
  - How sparse is the incidence matrix?

Boolean Retrieval

Methodology

- Our goal is **to develop a system to address the ad hoc retrieval task**
  - a system aims to provide documents
  - from within the collection that are relevant to an arbitrary user information need
  - communicated to the system by means of a one-off, user-initiated query
- What is a document?
  - a single page?
  - a book?
  - a paragraph?
  - examples, Wikipedia pages, news articles, tweets, news events
  - introduce **corpus** and **documents**
- **Query** and **Information Need** is not synonymous

Boolean Retrieval

Information Need

- An information need is the topic about which the user desires to know more
- Query is the user generated input to represent the information need
  - Attempt to communicate the information need
  - Difference in communication and understanding
- A document is **relevant**
  - if it is one that the user perceives as containing information of value
  - with respect to their personal information need
- To assess the effectiveness of an IR system
  - Precision
    - What fraction of the returned results are relevant to the information need?
  - Recall
    - What fraction of the relevant documents in the collection were returned by the system?

Boolean Retrieval

Effectiveness Understanding

- Which one would you prefer and when ?
  - You are building an automated tool for **Disaster Management** that aids government agencies to identify
  - **Resource** needs, **healthcare** requirement, **affected** populations, etc.
  - if a trade off, would you accept some form of **irrelevant** information?
  - Or, not lose any **relevant** information?

Representation

- Which one would you prefer and when ?
  - Flagging a news as **fake news**
  - If a trade off, would you accept some form of **fake** news?
  - Or, not lose any **non-fake** news?

Boolean Retrieval

Inverted Index

- A term-document matrix in a naive way
  - has high sparsity
- A document that seems **highly** similar to a query might be just because of the data
- Requires huge space and needs lots of efforts to filter out
  - unnecessary information, that is trivial and intuitive
- Formulate an **inverted index**
  - keep a dictionary of terms, i.e., vocabulary
  - for each term/lexicon/token
    - have a list that records which documents the term occurs in
    - referred as **Postings List**
- Handle sparsity
- Ensures more importance to if we co-occur together
  - Rather than absent together

Boolean Retrieval

Inverted Index

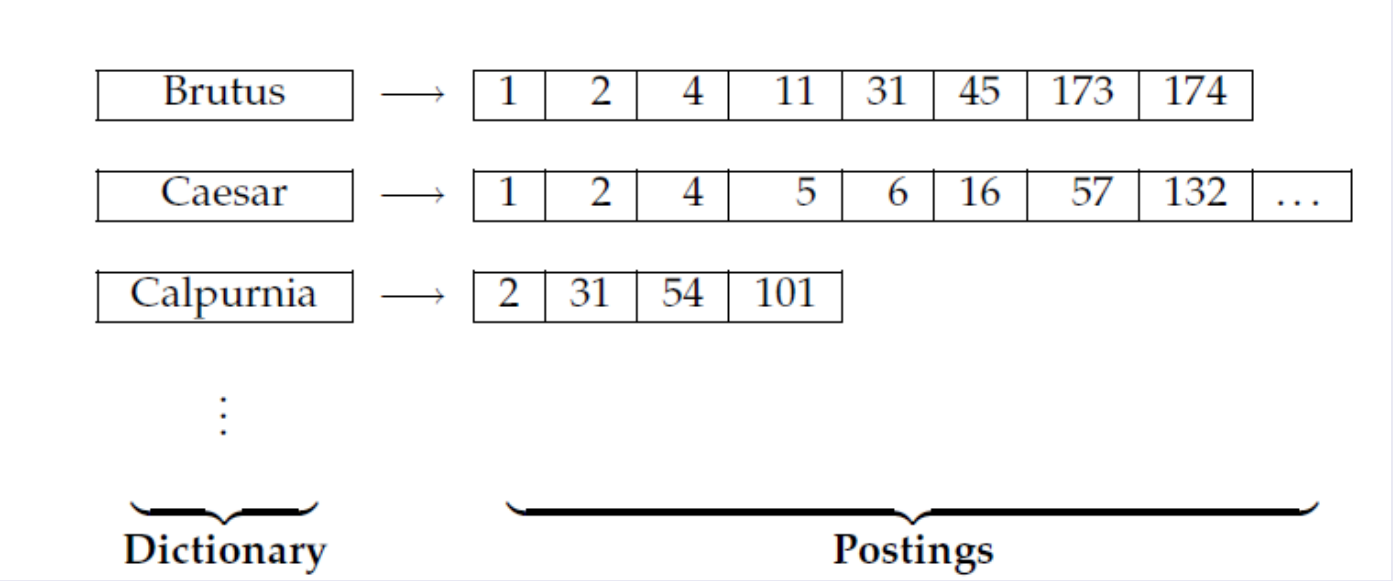
- Sorted alphabetically for quicker access
- Document List sorted increasing order

Steps to Follow

- Tokenize the articles and then, sentences
- Do linguistic **pre-processing**
  - producing a list of **normalized** tokens
- For every token
  - Identify the documents it appeared in
  - Store that information in form of linked list/ any other data structure
- Index the documents that each term occurs in
  - create an **inverted index**
  - represents a **dictionary** and **postings**

Boolean Retrieval

Inverted Index



Boolean Retrieval

Pre-processing Details

- Is every document equally relevant if it just occurs?
  - Consider you are retrieving all news articles than mention *Kohli* on last 10 days
  - You found around 200 news articles, so all of them are retrieved
  - Precision is ——————- ?
- Very few, say $k\%$ of these news articles, talk about *Women WC* and *Smriti Mandhana*
  - Compared with *Kohli*, so are they as relevant as other news articles?
  - What could have been an issue?
- Presence does not equate with *relevance* always
  - Specially, false positives
  - Frequency is a better metric
  - Other forms are also present

Pre-processing Details

Stop-words Removal

- Is every word equally important?
  - Considering all words have increased the size of our incidence matrix
  - Some words are hardly, **relevant**, say **stop-words**
  - Extremely common words
    - appear to be of little value in helping select documents
    - increase generic information
- What is a stop-word?
  - Frequency based analysis
  - High frequency, equates with **generic** information
  - **Relevance** requires **specific** information
- How to find a stop word?
  - determine a stop list is to sort the terms by collection frequency
  - the total number of times each term appears in the document collection)

Pre-processing Details

Stop-words Removal

a	an	and	are	as	at	be	by	for	from
has	he	in	is	it	its	of	on	that	the
to	was	were	will	with					

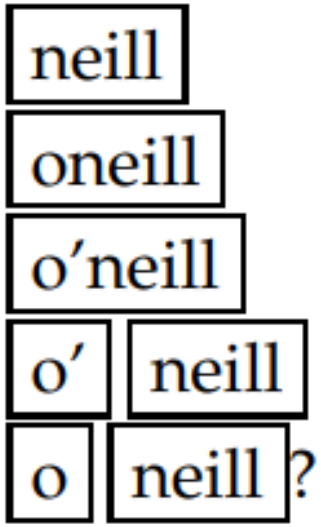
Pre-processing Details

- Remove by pre-processing
  - What about phrases?
  - PM, PMO, Prime Minister of India, Prime Minister and Home Minister
  - *To be or not to be*

Pre-processing Details

Tokenization

- *Mr. O'Neill thinks that the boys' stories about Chile's capital aren't amusing*
- Also, issues of tokenization are language-specific



Pre-processing Details

Compound Words

- Compound nouns without Spaces
  - *Computerlinguistik* 'computational linguistics'
  - *Lebensversicherungsgesellschaftsangestellter* means life insurance company employee'
  - Seeing if a word can be subdivided into multiple words that appear in a vocabulary

Pre-processing Details

Normalization

- If you search for USA,
  - you might hope to also match documents containing U.S.A
  - However, if I put in as my query term **C.A.T.**,
    - if it finds all **cat** in documents
- *anti-discriminatory* and *antidiscriminatory*
- *car* and *automobile*
- Accents and diacritics

Pre-processing Details

Stemming and lemmatization

- *organize*, *organizes*, and *organizing*
  - *democracy*, *democratic*, and *democratization*
- Reduce inflectional forms/ related forms of a word
  - to a common base form
  - am, are, is **normalizes to** be
  - car, cars, car's, cars' **normalizes to** car
- Stemming is a crude heuristic process
  - chops off the ends of words
- Lemmatization usually refers to doing things
  - properly with the use of a vocabulary
  - morphological analysis of words

Pre-processing Details

Solutions

- Index unnormalized tokens
  - maintain a query expansion list
  - expand the user generated query
  - data arrangement has every separate entry
  - query is then effectively a disjunction of several postings lists
- Index normalized tokens
  - the document contains automobile, we index it under car
  - either of these methods is considerably less efficient

Thank you!